

	At	CPU	Quantum=1ms	Priority		
P0	4	1		0		
P1	3	7		2		
P2	3	2		1		

FCFS

	P1	P2	P0				
0	3	10	12	13			

As there are no processes at instance 0

As p1 and P2 coming at same time –consider the subscript P1 is considered with 7 msec

Consider P2 with cpu time 2msec

Last process p0 at end

WT of P0

Start time –arrival time

12- 4 =8 msec

TAT

Completion time – arrival Time

P0

13-4 =9msec

Waiting time of P0 + its cpu time

8msec+1 cpu time =9 msec

	At	CPU	Quantum=1ms	Priority		
P0	4	1		0		
P1	3	6		2		
P2	3	2		1		

	P1	P2	P0	P1	P2	P1	P1	P1	P1	
0	3	4	5	6	7	8	9	10	11	12

Round Robin

Similar to FCFS but equal quantum is assigned to all processes

WT of P1

$$3-3 + (6-4) + (8-7) = 3\text{msec}$$

TAT p1

$$12-3 = 9\text{msec}$$

	At	CPU	Quantum=1ms	Priority		
P0	4	1		0		
P1	3	4		2		
P2	3	3 rem 2		1		
P3	0	9 rem 6		2		

SJF

First consider arrival processes(Preemptive SJF)

P3	P2	P0	P2	P1	P3	
0	3	4	5	7	11	17

Non preemptive SJF

The process after allocation of CPU, it completes till end

P3	P0	P2	P1			
0	9					

Priority

Consider arrival time

And priority

Case 1 = higher priority value is least priority process

Case 2 = lowest priority value is least priority process

	At	CPU	Quantum=1ms	Priority		
P0	4	1		0		
P1	3	4		2		
P2	3	3 rem=2		1		
P3	0	9 rem =6		2		

Case 1:

Low priority value =highest priority

P3	P2	P0	P2	P1	P3		
0	3	4	5	7	11	17	

WT of P3 $0-0+(11-3) = 8\text{msec}$

	At	CPU	Quantum=1ms	Priority		
P0	4	1		0		
P1	3	4		2		
P2	3	3		1		
P3	0	9		2		

Case 2:

Low priority value =low priority (non pre-emptivepriority)

P3	P1	P2	P0				
0	9	13	16	17			

Low priority value =low priority (preemptivepriority)

As the process P3 still having highest priority – it will continue

During tie in priority no matter what remaining cpu of other processes is less than current process, continue with current to avoid context switch

P3	P1	P2	P0	
0	9	13	16	17

3 queues (1level RR=4,2level RR 8quantum, last level FCFS)

1level RR =4quantum

2level round robin Quantum = 8 msec

3 level FCFS

Let us say Process P0 comes with cpu time 15msec

In 1st level Process executes for 4 msec only

Switches to 2nd level where it executes another 8 msec (overall exe =12msec out of 15msec)

It has to go to the last level(3 level) where it executes remaining 3ms scheduled as FCFS